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import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix o
import matplotlib.pyplot as plt # Import Matplotlib to visualize the out

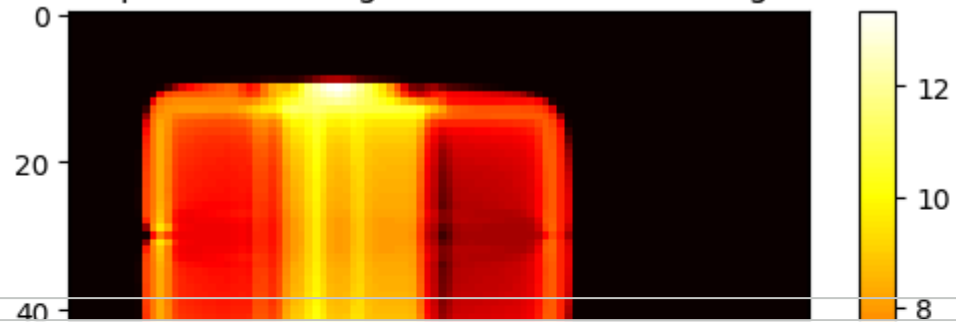
# Step 1: Create two sequential frames simulating an object moving to the right
frame1 = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.rectangle(frame1, (20, 20), (40, 40), 255, -1) # Draw a filled or outlined rectangle on the imag

frame2 = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.rectangle(frame2, (40, 20), (60, 40), 255, -1) # Object moved +20 pixels

# Step 2: Compute Dense Optical Flow (A core surveillance algorithm)
# This calculates the velocity vector of every pixel between the two frames
flow = cv2.calcOpticalFlowFarneback(frame1, frame2, None, 0.5, 3, 15, 3, 5, 1.2, 0) # Compute dense
mag, ang = cv2.cartToPolar(flow[..., 0], flow[..., 1]) # Convert Cartesian (x,y) vectors to polar (m

# Step 3: Visualize the motion magnitude
plt.imshow(mag, cmap='hot') # Render the image array to the display
plt.title('Optical Flow Magnitude (Motion Tracking)') # Add a descriptive title to the plot
plt.colorbar() # Add a color legend bar to the plot
plt.show() # Show the final plot window to the user
```

Optical Flow Magnitude (Motion Tracking)



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